

This file includes the: (1) Source Code and (2) Program Output.

Source Code

.java file link: [BasicATMSystemKarim](#)

```
import java.util.Scanner;

public class BasicATMSystemKarim {

    // output: menu
    public static void menu() {
        System.out.println("==== ATM MENU =====");
        System.out.println("(1) Check Balance");
        System.out.println("(2) Deposit");
        System.out.println("(3) Withdraw");
        System.out.println("(4) Exit");
    }

    // process: deposit
    public static double deposit(double balance, double amount) {
        return balance + amount;
    }

    // process: withdraw
    public static double withdraw(double balance, double amount) {
        return balance - amount;
    }

    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);

        double balance = 5000;
        int choice;

        do {
            menu();
```

```
// validate menu choice
while (true) {
    System.out.print("Enter your choice: ");
    if (input.hasNextInt()) {
        choice = input.nextInt();
        if (choice >= 1 && choice <= 4) {
            break;
        } else {
            System.out.println("Invalid choice.");
        }
    } else {
        System.out.println("Invalid input.");
        input.next();
    }
}

switch(choice) {

    case 1:
        System.out.println("Current Balance: " +
balance);

        break;

    case 2:
        double depositAmount;

        // loop until valid deposit
        while (true) {
            System.out.print("Enter amount to deposit:
");

            if (input.hasNextDouble()) {
                depositAmount = input.nextDouble();

                if (depositAmount > 0) {
                    break;
                } else {
```

```
                System.out.println("Amount must be
more than 0.");
            }

        } else {
            System.out.println("Invalid amount.");
            input.next();
        }
    }

    balance = deposit(balance, depositAmount);
    System.out.println("[ Deposit successful! ]");
    break;

case 3:
    double withdrawAmount;

    // loop until valid withdraw
    while (true) {
        System.out.print("Enter amount to withdraw:
");

        if (input.hasNextDouble()) {
            withdrawAmount = input.nextDouble();

            if (withdrawAmount <= 0) {
                System.out.println("Amount must be
more than 0.");

            } else if (withdrawAmount > 2000) {
                System.out.println("Oops! Maximum
withdrawal is 2000 only.");
            } else if (withdrawAmount > balance) {
                System.out.println("Sorry!
Insufficient balance.");
            } else {
                break;
            }
        }
    }
}
```

```
        } else {
            System.out.println("Invalid amount.");
            input.next();
        }
    }

    balance = withdraw(balance, withdrawAmount);
    System.out.println("[ Withdrawal successful! ]");
    break;

    case 4:
        System.out.println("[ Thank you for using the
ATM! See you again! ]");
        break;
    }

    System.out.println();

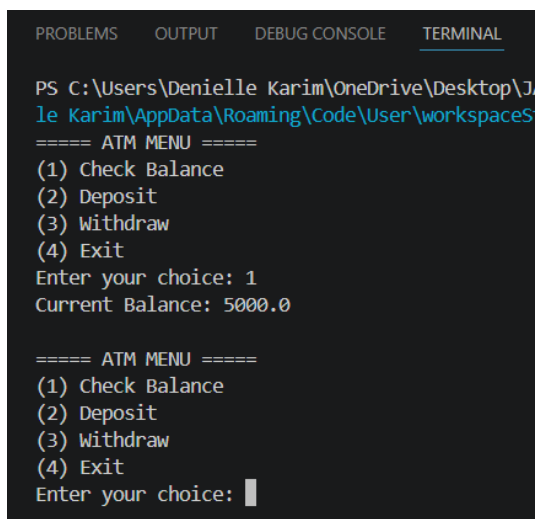
    } while(choice != 4);

    input.close();
}
}
```

Program Output

(1) Valid choice in Menu

- Proceeds to Menu once successful

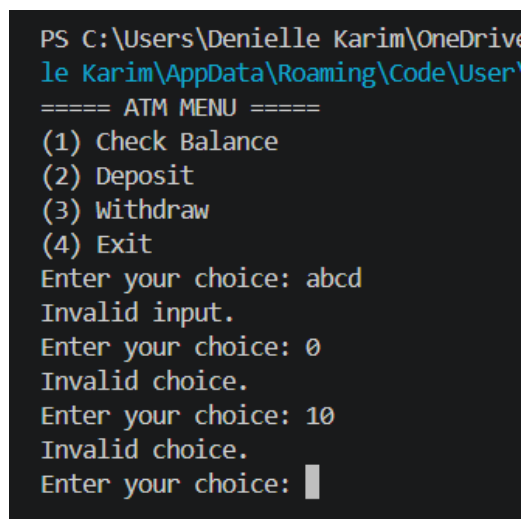


```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL
PS C:\Users\Denielle Karim\OneDrive\Desktop\J...
le Karim\AppData\Roaming\Code\User\workspaces
===== ATM MENU =====
(1) Check Balance
(2) Deposit
(3) Withdraw
(4) Exit
Enter your choice: 1
Current Balance: 5000.0

===== ATM MENU =====
(1) Check Balance
(2) Deposit
(3) Withdraw
(4) Exit
Enter your choice: █
```

(2) Invalid choice in Menu

- Loops until it gets a valid input



```
PS C:\Users\Denielle Karim\OneDrive\Desktop\J...
le Karim\AppData\Roaming\Code\User\workspaces
===== ATM MENU =====
(1) Check Balance
(2) Deposit
(3) Withdraw
(4) Exit
Enter your choice: abcd
Invalid input.
Enter your choice: 0
Invalid choice.
Enter your choice: 10
Invalid choice.
Enter your choice: █
```

(3) Checking balance

```
PS C:\Users\Denielle Karim\OneDrive\le Karim\AppData\Roaming\Code\User
===== ATM MENU =====
(1) Check Balance
(2) Deposit
(3) Withdraw
(4) Exit
Enter your choice: 1
Current Balance: 5000.0
```

(4) Exiting the Menu

```
===== ATM MENU =====
(1) Check Balance
(2) Deposit
(3) Withdraw
(4) Exit
Enter your choice: 4
[ Thank you for using the ATM! See you again! ]

PS C:\Users\Denielle Karim\OneDrive\Desktop\JAVA VSCODE
```

(4) Valid Deposit amount
- Shows a successful message

```
PS C:\Users\Denielle Karim\OneDrive\le Karim\AppData\Roaming\Code\User
===== ATM MENU =====
(1) Check Balance
(2) Deposit
(3) Withdraw
(4) Exit
Enter your choice: 2
Enter amount to deposit: 500
[ Deposit successful! ]
```

(5) Invalid Deposit amount
- Loops until it gets valid amount

```
PS C:\Users\Denielle Karim\OneDrive\le Karim\AppData\Roaming\Code\User
===== ATM MENU =====
(1) Check Balance
(2) Deposit
(3) Withdraw
(4) Exit
Enter your choice: 2
Enter amount to deposit: abcd
Invalid amount.
Enter amount to deposit: 0
Amount must be more than 0.
Enter amount to deposit: -90
Amount must be more than 0.
Enter amount to deposit: █
```

(6) Valid Withdrawal amount
- Shows a successful message
and proceeds to Menu

```
PS C:\Users\Denielle Karim\OneDrive\Desktop\le Karim\AppData\Roaming\Code\User\works
===== ATM MENU =====
(1) Check Balance
(2) Deposit
(3) Withdraw
(4) Exit
Enter your choice: 3
Enter amount to withdraw: 500
[ Withdrawal successful! ]

===== ATM MENU =====
(1) Check Balance
(2) Deposit
(3) Withdraw
(4) Exit
Enter your choice: █
```

(7) Invalid Withdrawal amount
- Loops until it gets valid amount

```
===== ATM MENU =====
(1) Check Balance
(2) Deposit
(3) Withdraw
(4) Exit
Enter your choice: 3
Enter amount to withdraw: abcd
Invalid amount.
Enter amount to withdraw: 1500
Sorry! Insufficient balance.
Enter amount to withdraw: █

PS C:\Users\Denielle Karim\OneDrive\Desktop\le Karim\AppData\Roaming\Code\User\works
===== ATM MENU =====
(1) Check Balance
(2) Deposit
(3) Withdraw
(4) Exit
Enter your choice: 3
Enter amount to withdraw: 2500
Oops! Maximum withdrawal is 2000 only.
Enter amount to withdraw: █
```